

TEZPUR UNIVERSITY
End Semester Test, Autumn 2017
CO 504 Natural Language Processing

Marks : 60

Time : 2 hours

The figures in the right-hand margin indicate marks for the individual questions.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. How is speech converted to digital form? How is digital speech different from IPA transcription of expressions?
3 + 2
2. Draw a *finite state transducer* for morphological parsing of the words *walked*, *countries* and *wrote*. 4
3. (a) How can the concept of *n-grams* be used in syntax analysis? Why is a *balanced* corpus desirable? 3 + 1
(b) Write a CFG that depicts the ambiguity of the sentence *This is for old men and women*. 3
4. (a) How would *features and unification* help in syntax analysis of the following sentences- 3
(i) The boys is strong. (ii) The boys are strong.
(b) What are some drawbacks of *probabilistic context free grammar*? 3
5. What are the types of *quantifiers* in First Order Predicate Calculus? With examples explain how quantifiers with *reification* are useful? 2 + 3

6. (a) Identify synonymy, homonymy, homophones, homographs, hyponymy and polysemy among the following words (some hints of meaning are provided within brackets)- 6

write, read (present tense), right (true), act (pretend as someone else), react (take steps in response to a situation), red, car, read (past tense), vehicle, correct (adjective), act (take expected step), react (action when certain chemicals are mixed).

7. How can prepositions and suffixes help in determining *thematic roles* of arguments of verbs in some languages? 3

8. (a) Using examples illustrate how *word sense ambiguity* is distinct from *structural ambiguity* of a sentence? 3

(b) Suggest how the lexicon entries of ambiguous words can be used to attempt disambiguation of such words in a given expression? 3

9. (a) How do occurrence count of a query term in a document, and count of documents in which the term occurs affect its weight in information retrieval? 3

(b) How can homonymy affect information retrieval? 2

10. (a) Identify the *referring expressions*, *referents* and a *cataphora* in the following discourse. State how pronoun resolution can be achieved in the passage. 3 + 3

When he went to buy a car, Ravi had already obtained a loan. His wife Mina preferred a small car. She wanted it to be fuel efficient. They finally chose an Alto.

- (b) What are some common *coherence relations*. Give examples. 3

11. (a) What is *lexical gap* in the context of translation? 2

(b) Despite being a difficult problem, how is good quality machine translation possible in certain domains? 2